Glimpses of SONiC India Workshop 2025

the rapidly increasing interest and engagement within the SONiC community.

The workshop was inaugurated by **Dave Maltz**, Microsoft's Distinguished CVP and chair of the Governing Board of SONiC community, and **Lihua Yuan**, Microsoft's VP for Engineering and chair of SONiC's Technical Steering Committee. In his keynote, Dave emphasized the importance of the SONiC community, noting its remarkable growth over the years and its pivotal role in enabling **AI-ready networking ecosystems**. Lihua followed by highlighting Microsoft's ongoing contributions, outlining recent **advancements in SONiC** and the company's vision for building scalable, efficient, and intelligent networks.

The technical sessions showcased the breadth of SONiC's impact across industries. Broadcom presented hardware innovations for architecting scalable, lossless AI/HPC networks, showcasing its comprehensive portfolio for scale-up and scale-out deployments, with particular emphasis on the Tomahawk6/Ultra and Jericho4 chips. Presenters from Upscale AI shared insights on using SONiC as a management plane in UA Link, underscoring the need for an open, vendor-neutral standard to meet AI workloads' unprecedented demands for interconnect bandwidth and low latency. Arista Networks presented their work on Layer-1 telemetry and troubleshooting in SONiC, especially with respect to FEC (Forward Error Correction) and FLR (Frame Loss Ratio), illustrating how SONiC enables more effective monitoring and issue resolution in complex environments.

NPCI, whose network runs on ~50% SONiC devices, showcased one of SONiC's most practical and impactful use cases—building India's payment backbone. With a vision to adopt open source frameworks, NPCI leveraged SONiC to architect a robust, scalable network infrastructure capable of handling the staggering load of trillions of transactions per day, underscoring SONiC's maturity and reliability in mission-critical environments. **BE Networks**, a network orchestration and management service provider, shared their work on addressing the challenges of **lifecycle management in SONiC NOS switching fabrics**, including maintenance, upgrades, and scalability, with a focus on enabling robust **Day-2+ operations. Dell** showcased its contributions towards **AI networking using SONiC**, highlighting innovations such as **linear pluggable optics with SONiC, SmartFabric Manager integration**, and deployment services tailored for AI-driven infrastructures. **Marvell Networks** offered insights into preparing SONiC for **large-scale, scale-out networks,** emphasizing the importance of features like **Link Layer Recovery** and **Credit-Based Flow Control** to ensure reliability and performance at cloud scale.